5.3 Asexual reproduction

We saw that plants reproduce not only from seeds but by other processes as well. The production of new plants without the involvement of pollination and fertilization is known as asexual reproduction. Let us study the types of asexual reproduction.

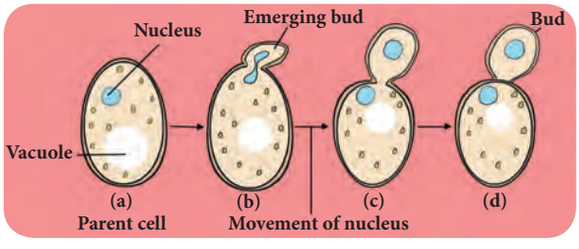
5.3.1 Vegetative Propagation

In potato, shoot arise from eyes. Sugar cane and yam also grow like this. Vegetative parts of the plants such as root, stem and leaves can help to produce the plant.



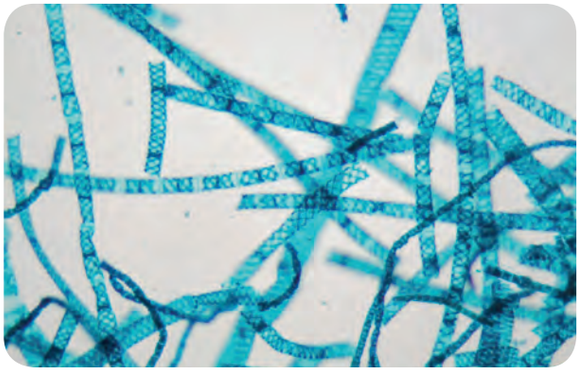
5.3.2 Budding

When we go to a bakery we see so many types of cakes and breads. These are very soft in nature. This is due to the presence of yeast. Single yeast undergoes asymmetric division. It produces a small protuberance which gradually grow and detach from the parent cell. This process is called **budding**.



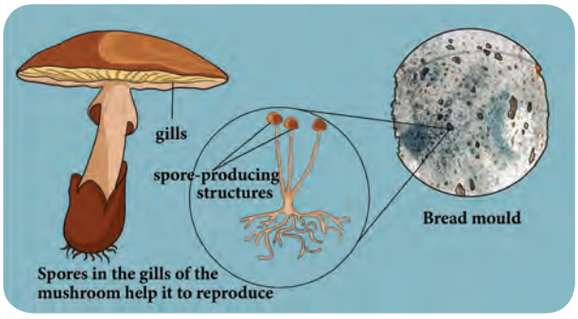
5.3.3 Fragmentation

In a pond we see so many algae. **Spirogyra** is a filamentous alga. When it matures, the filament divides into pieces. Each fragment or piece of a filament will grow into a new filament or individual. Likewise **spirogyra** produces so many young ones and this process is known as **fragmentation.**



5.3.4 Spore Formation

Scarcity of water, high temperature, nutrient deficiency in soil etc., are unfavourable conditions. During these conditions non-flowering plants like algae, fungi, moss and ferns produce spores. They germinate into a new plant when favourable conditions return.



5.4 Modifications of plant parts

Compare the given plants and discuss with your teacher.



Carrot Plant



Grass

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